

Chemistry Unit Test

Subject: Chemistry | Grade: 10th | Student: Fixture

Q1. [5 marks]

Balance the following chemical equation: $\text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{O}$

Answer:

$2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$. This is balanced: there are 4 hydrogen atoms and 2 oxygen atoms on both sides.

Q2. [4 marks]

Balance the following chemical equation: $\text{N}_2 + \text{H}_2 \rightarrow \text{NH}_3$

Answer:

$\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$. This is balanced because the nitrogen atoms match on both sides.

Q3. [5 marks]

The reaction $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca}(\text{OH})_2$ releases a large amount of heat. State whether this reaction is exothermic or endothermic, and explain in terms of energy absorbed or released.

Answer:

This reaction is exothermic. During an exothermic reaction, energy is absorbed from the surroundings, which is why heat is given off.

Q4. [5 marks]

Balance the following chemical equation: $\text{Zn} + \text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$

Answer:

$\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$. This equation is already balanced.

Q5. [6 marks]

Magnesium ribbon burns in oxygen to form a white powder. Name the product formed and write the balanced chemical equation for this reaction.

Answer:

The product is magnesium oxide (MgO), a white powder. The balanced equation is: $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$. Magnesium and oxygen atoms are balanced on both sides (2 Mg and 2 O).